Re: RFP No. 978-2025 – Feasibility Study for Graham Avenue District Thermal Energy System

The City of Winnipeg
510 Main Street
Winnipeg, MB R3B 1B9

Dear Sir or Madam,

GEOptimize Inc. is pleased to submit this proposal in response to RFP No. 978-2025 for the Feasibility Study for the Graham Avenue District Thermal Energy System. As a Winnipeg-based professional engineering firm specializing in ground source heat pump systems and district geothermal energy networks, we welcome the opportunity to support the City of Winnipeg in advancing a low-carbon district energy initiative aligned with the City's Climate Action Plan.

Founded in Manitoba and headquartered in Winnipeg, GEOptimize brings deep local knowledge of Manitoba's geology, climate, regulatory environment, and construction market. Our team has delivered district geothermal feasibility studies, advanced energy modelling, and commercial heat pump system designs for projects in Winnipeg and across North America.

For the Graham Avenue study, our approach integrates 8760-hour building energy modelling, transient district system simulation using TRNSYS, and conceptual geothermal system design to produce defensible projections of system performance, greenhouse gas reductions, and lifecycle economics. Our methodology is structured to meet the deliverables in RFP Section D4.

GEOptimize Inc. offers to perform the requested services for the fees indicated, subject to negotiation of a mutually agreeable contract. This proposal is valid for acceptance for ninety (90) days from the submission deadline.

Sincerely,

Ed Lohrenz, BES, CGD
Founder
GEOptimize Inc.
167 Lombard Avenue, Suite 500
Winnipeg, MB R3B 0V3
Phone: 204-996-8133
Email: ed@geoptimize.ca

Connor Dacquay, Dr. Eng, P.Eng, CGD
Engineer
GEOptimize Inc.

Ian Lohrenz, P.Eng, CGD
Engineer
GEOptimize Inc.

Feasibility Study for Graham Avenue District Thermal Energy System

Proposal For:

The City of Winnipeg
510 Main St
Winnipeg MB, R3B 1B9
Canada

Proposal Created By:

GEOptimize Inc.
167 Lombard Avenue,
Suite 500
Winnipeg, MB R3B 0V3
ed@geoptimize.ca
1 (204) 996 8133

Proposal Date:

February 12, 2026

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Section A – Form A: Proposal

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FORM A: BID/PROPOSAL
(See "Bid/Proposal" clause in Tender/RFP)

1. Contract Title

FEASIBILITY STUDY FOR GRAHAM AVENUE DISTRICT THERMALENERGY SYSTEM

2. Bidder/Proponent

GEOptimize Inc
Name of Bidder/Proponent

Usual Business Name of Bidder/ Proponent as it appears on Invoice (if different from above)

167 Lombard Ave, unit 500
Street

City	Province	Postal Code
Winnipeg	MB	R3B0V

ed@geoptimize.ca
Email Address of Bidder/Proponent

Facsimile Number

(Mailing address if different)

Street or P.O. Box

City Province Postal Code

GST Registration Number (if applicable)

(Choose one)

The Bidder/Proponent is:

- ☐ a sole proprietor
- ☐ a partnership
- ☒ a corporation

carrying on business under the above name.

3. Contact Person

The Bidder/Proponent hereby authorizes the following contact person to represent the Bidder/Proponent for purposes of the Bid/Proposal.

Contact Person	Title
Ed Lohrenz	Founder

Telephone Number	Facsimile Number
204-996-8133	

4. Definitions

All capitalized terms used in the Contract Documents shall have the meanings ascribed to them in the General Conditions and Tender/RFP documents.

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5.	Offer	The Bidder/Proponent hereby offers to perform the Work in accordance with the Contract for the price bid~~,~~ in Canadian funds, set out in the Bid/Proposal.
6.	Bid Security	Where Bid Security is required, the Bidder/Proponent provides bid security in the form of a bid bond (Form G1: Bid Bond and Agreement to Bond) in accordance with the Bid Security clause in the Tender document and agrees that it shall be held by the City in accordance with the Contract.
7.	Execution of Contract Documents	If required pursuant to C4, the Bidder/Proponent agrees to execute and return the Contract Documents no later than seven (7) Calendar Days after receipt of the Contract Documents, in the manner specified in C4.
8.	Commencement of the Work	The Bidder/Proponent agrees that no Work shall commence until he/she is in receipt of a notice of award from the Award Authority authorizing the commencement of the Work.
9.	Contract	By submitting a Bid/Proposal in response to this Tender/RFP, the Bidder/Proponent certifies that it has read, understands, and agrees to the terms and conditions of this Tender/RFP and that the Tender/RFP, in its entirety shall be deemed to be incorporated in and to form a part of this offer notwithstanding that not all parts thereof are necessarily attached to or accompany the Tender/Proposal.
10.	Addenda	The Bidder/Proponent certifies that the following addenda have been received and agrees that they shall be deemed to form a part of the Contract:

No.	Dated

11.	Time	This offer shall be open for acceptance, binding and irrevocable for a period of Ninety (90) Calendar Days following the Submission Deadline.
12.	Indigenous Self-Declaration	The City is requesting that Bidders/Proponents identify if their business is at least 51% owned by one or more Indigenous persons of Canada. ☐ YES, 51% or more Indigenous ownership ☒ NO, it is not This information is being gathered for statistical purposes only and will not be used for purposes of evaluation.

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13. Signatures

The Bidder/Proponent or the Bidder's/Proponent's authorized official or officials have signed this

1st day of February, 20 26 .

Signature of Bidder/ Proponent or
Bidder's/Proponent's Authorized Official or Officials

(Print here name and official capacity of individual whose signature appears above)

Ed Lohrenz, Founder

(Print here name and official capacity of individual whose signature appears above)

Section B – Fees

TASK	HOURS			
Phase 1: Project Start-Up Meeting & Information Gathering	**Stephen Hak**	**Ian Lohrenz**	**Connor Dacquay**	**Ed Lohrenz**
Kick-off meeting	2			1
Information gathering	8			
	$ 2,250	$ -	$ -	$ 250
Phase 2: Desktop Geotechnical Study				
Reviewing available subsurfance information for the study area	20			4
Development of geotechnical report	16	16	16	4
	$ 8,100	$ 3,600	$ 3,600	$ 2,000
Phase 3: Preliminary Progress Report & Review				
Preliminary 8760 building energy models development		450		20
Preliminary district design & option analysis	60		60	20
	$ 13,500	$ 101,250	$ 13,500	$ 10,000
Phase 4: Conceptual Design & Energy Modelling				
Refine 8760 building energy models		255		10
Conceptual district system design using feedback from stakeholders	20		10	10
	$ 4,500	$ 57,375	$ 2,250	$ 5,000
Phase 5: 75% Draft Feasibility Report & Review				
Economic & energy use analysis including Class C pricing	40	20	20	8
Pricing partners	48			
Report writing	40	20	20	40
	$ 28,800	$ 9,000	$ 9,000	$ 12,000
Phase 6: Final Feasibility Report & Presentation				
Final report writing	40	20	20	16
Final presentation	20	10	10	8
	$ 13,500	$ 6,750	$ 6,750	$ 6,000
TOTAL	$			318,975.00

HOURLY RATES			
Stephen Hak	Ian Lohrenz	Connor Dacquay	Ed Lohrenz
$ 225.00	$ 225.00	$ 225.00	$ 250.00

PERCENTAGE OF TOTAL WORK			
Stephen Hak	Ian Lohrenz	Connor Dacquay	Ed Lohrenz
22%	56%	11%	10%

Section C – Experience of Proponent and Subconsultants

Firm Overview

GEOptimize is a Manitoba-based professional engineering firm specializing exclusively in the planning, modelling, design, and optimization of ground source heat pump (GSHP) and district-scale geothermal energy systems. Founded in 2013, GEOptimize supports public and private-sector clients as they transition away from fossil fuel–based heating and cooling systems toward high-efficiency, low-carbon thermal energy solutions.

The firm's practice is focused on both single-building and multi-building district applications, including feasibility studies, conceptual system design, design assistance, energy modelling, detailed engineering, forensic analyses, lifecycle cost analysis, and system optimization. GEOptimize's work emphasizes the integration of thermal energy networks (TENs), energy sharing between buildings, and long-term system performance informed by realistic operational assumptions.

GEOptimize maintains in-house expertise in hourly building energy modelling, ground heat exchanger (GHX) design, and district system simulation. Unlike general mechanical engineering firms that evaluate geothermal as one of many services, GEOptimize specializes exclusively in geothermal and thermal energy network systems. The firm employs six staff and uses commercially available, industry-recognized software tools and custom modelling approaches to evaluate technical feasibility, infrastructure requirements, and long-term performance and risk. This capability enables GEOptimize to support large feasibility studies and defensible business case development.

GEOptimize is licensed to provide professional engineering services in the Province of Manitoba and Ontario. GEOptimize routinely works collaboratively with multidisciplinary project teams, municipal stakeholders, utilities, and regulatory bodies.

Local District GSHP System Experience

Water Tower District – Winnipeg, Manitoba – 2025

Project Description
GEOptimize completed a feasibility study in 2025 for a large-scale district geothermal system serving a proposed mixed-use development by Shindico Realty Inc. and Oxala Developments in Winnipeg. The project encompasses approximately 165 acres and includes 38 buildings totaling approximately 68,000 m^2 (733,000 ft^2) of residential, commercial, retail, and industrial space.

The objective of the study was to determine the technical and economic feasibility of implementing a ground source heat pump (GSHP)–based thermal energy network (TEN) to provide space heating, cooling, and domestic hot water in place of fossil fuels and electric resistance heating. The TEN was evaluated as a utility-owned and operated geothermal utility regulated by the Public Utilities Board.

Multiple geothermal source configurations were assessed, including open-loop groundwater systems utilizing extraction and reinjection wells, closed-loop horizontal ground heat exchangers, closed-loop vertical borehole systems, and hybrid configurations integrating open- and closed-loop systems.

The study also evaluated opportunities to integrate waste heat sources and wastewater heat recovery to improve overall system efficiency and reduce greenhouse gas emissions.

A key component of the feasibility assessment involved evaluating subsurface conditions. Historical site usage indicated the presence of a productive aquifer, suggesting that partial or full reliance on groundwater-based systems could significantly reduce capital costs relative to fully closed-loop systems. The analysis demonstrated how even partial utilization of groundwater resources could materially reduce ground heat exchanger infrastructure requirements and improve overall project economics.

Role of GEOptimize
GEOptimize was responsible for the full technical and economic feasibility assessment. This included development of building energy models based on developer-provided information, district system conceptual design, geothermal source evaluation, and lifecycle performance analysis. The team used advanced simulation tools to optimize system configuration and long-term thermal balance, evaluated alternative geothermal source options, assessed aquifer feasibility, and prepared the technical and financial feasibility analysis supporting a utility-based ownership and operating model.

Project Cost
Original Contracted Cost: CAD 162,000
Final Cost: CAD 162,000 (completed on budget)

Design and Schedule
Anticipated Design Schedule: 6 months
Actual Design Schedule: 8 months (extended due to delays in obtaining building and site information)

Project Owner
Shindico Realty Inc. and Oxala Developments

Reference Information
Robert Scaletta
Shindico Realty Inc.
Phone: 204-928-8242
Email: rscaletta@shindico.com

Jana Brunel
Efficiency Manitoba
Phone: 204-390-1054
Email: jana.brunel@efficiencymb.ca

Canadian Mennonite University – Winnipeg, Manitoba – 2023 to Present

Project Description

GEOptimize has been directly involved in the design and implementation of an operational district GSHP system at Canadian Mennonite University (CMU) in Winnipeg since 2023. The CMU campus consists of nine academic and residential buildings, with the district geothermal system developed to replace aging mechanical systems and support the University's long-term decarbonization objectives.

The district GSHP system enables energy sharing between campus buildings and is designed with a phased, modular approach, allowing additional buildings to be connected over time as budgets permit. The system intersperses ground heat exchangers with connected buildings along a central ambient thermal loop, enabling buildings to add or remove heat as required while maintaining stable loop temperatures.

Key technical features include directionally drilled horizontal GHXs installed beneath constrained site areas, approximately 16 km of GHX piping installed in the initial phases (expanding to an anticipated 40 km as the system is fully built out), centralized loop infrastructure housed in a compact, serviceable enclosure containing manifolds, circulation pumps, controls, and monitoring equipment, and a phased connection strategy supporting incremental expansion and long-term system optimization.

Ongoing monitoring of the system informs operational adjustments and enables the system to be right-sized over time, reducing unnecessary upfront capital investment and improving long-term performance certainty. The CMU project demonstrates GEOptimize's ability to support the technical delivery, project management, and long-term operational evolution of district thermal energy systems.

Role of GEOptimize

GEOptimize led the geothermal system master planning, construction coordination, detailed design, and phased implementation of the district GSHP system. Responsibilities included development of the district system concept, GHX field design, ambient loop routing and sizing, building interface design, system controls strategy, and construction-phase technical support. GEOptimize team members also established the long-term monitoring framework and provides ongoing advisory support to optimize system performance and guide future expansion phases.

Project Cost
Original Contracted Cost: CAD ~ 2 million (for first phase)
Final Cost: CAD ~ 2 million

Design and Schedule
Anticipated Design Schedule: 12 months (for first phase)
Actual Design Schedule: 12 months

Project Owner
Canadian Mennonite University

Reference Information
Randy Neufeld
Canadian Mennonite University
Phone: 204-487-3300 ext. 324
Email: rneufeld@cmu.ca

Andrew Hamstra
Canadian Mennonite University
Email: ahamstra@cmu.ca

Other District GSHP System Experience

Municipality of the County of Inverness – Inverness, Nova Scotia – 2022

Project Description
GEOptimize completed a district geothermal feasibility study in 2022 for the Municipality of the County of Inverness to evaluate the conversion of multiple community buildings to a GSHP system. The study focused on six municipal and institutional facilities, including the Hospital, Manor, Arena, Education Centre–Academy, Co-op Inverness, and the Wastewater Treatment Plant.

The objective of the study was to determine the technical and economic feasibility of implementing a GSHP-based district energy sharing system to replace existing oil- and electricity-based heating, cooling, and domestic hot water systems.

The feasibility assessment included detailed building energy modeling, subsurface characterization, conceptual district system design, and lifecycle economic analysis. GEOptimize developed calibrated building energy models and a district system simulation using TRNSYS to assess long-term thermal performance, GHX sizing, and system interactions. Multiple geothermal source and heat recovery configurations were evaluated, including vertical boreholes, horizontal directionally drilled GHXs, groundwater systems, wastewater heat recovery, surface water heat exchangers, and coal mine water wells.

The preferred concept was an energy-sharing district system using horizontal directional GHXs with provisions for phased implementation and future expansion. The study also evaluated building-level retrofit requirements, including replacement of boilers and chillers with water-to-water heat pumps, low-temperature coil conversions, and waste heat recovery from refrigeration systems. Subsurface thermal properties were estimated from regional well logs to support preliminary GHX sizing and long-term performance projections.

Role of GEOptimize
GEOptimize served as the prime geothermal and energy systems consultant and co-author of the feasibility report. Responsibilities included development of detailed 8760-hour building energy models, district system simulation using TRNSYS, geothermal source evaluation, conceptual system design, subsurface analysis, and lifecycle economic evaluation.

Project Cost
Original Contracted Cost: CAD 85,000
Final Cost: CAD 85,000 (completed on budget)

Design and Schedule
Anticipated Design Schedule: 6 months
Actual Design Schedule: 6 months

Project Owner
Municipality of the County of Inverness

Reference Information
Peter Simpson
EastPoint Engineering Ltd.
Phone: 902-258-1111
Email: psimpson@invernesscounty.ns.ca

Andrew MacDonald
EastPoint Engineering Ltd.
Phone: 902-423-5555
Email: amacdonald@eastpointeng.ca

Bemidji State University – Bemidji, Minnesota – 2021

Project Description

GEOptimize completed a district GSHP feasibility study in 2021 for Bemidji State University (BSU) and Otter Tail Power Company. The study evaluated the conversion of 23 campus buildings totaling approximately 1.45 million square feet from a central natural-gas-fired steam and electric chiller system to a district GSHP system, supporting BSU's commitment to achieve carbon neutrality by 2050.

The feasibility assessment included development of detailed 8760-hour building energy models calibrated with historical utility consumption. Multiple configurations were evaluated, including horizontal directionally drilled ground heat exchangers, vertical boreholes, surface water heat exchangers in Lake Bemidji, and groundwater-based heat exchangers. The study also assessed complementary energy efficiency measures, including energy recovery ventilators and lighting upgrades, to reduce heating and cooling loads and optimize system sizing.

The recommended system concept utilized a district ambient-loop "energy transfer pipe" configuration, enabling energy sharing between buildings and modular, phased expansion. The preferred geothermal source consisted primarily of horizontal directional drilling (HDD) GHXs due to available land area and lower installation cost relative to vertical borefields. The study demonstrated that excluding efficiency measures would increase required GHX capacity and cost by approximately 40 percent, materially reducing project feasibility.

Role of GEOptimize

GEOptimize served as the prime consultant and was responsible for developing building energy models using Trane TRACE™ 700, calibrating them to historical utility data, performing GHX sizing, designing conceptual district system configurations, and completing the technical, economic, and environmental feasibility analysis. GEOptimize also evaluated phased implementation strategies and modular expansion options to minimize upfront capital investment and support long-term campus-wide decarbonization.

Project Cost

Original Contracted Cost: USD 56,000
Final Cost: USD 56,000

Design and Schedule

Anticipated Design Schedule: 8 months
Actual Design Schedule: 8 months

Project Owner

Bemidji State University (with Otter Tail Power Company as project partner)

Reference Information
Jon Fabre
Otter Tail Power Company
Phone: 218-739-8633
Email: jfabre@otpco.com

Erika Bailey-Johnson
Sustainability Director, Bemidji State University
Phone: 218-755-2560
Email: Erika.Bailey-Johnson@bemidjistate.edu

MRL Triangle District – Missoula, Montana – 2023

Project Description
GEOptimize completed a district GSHP feasibility study in 2023 for the Montana Rail Link (MRL) Triangle redevelopment site in Missoula, Montana. The 11.4-acre brownfield site is planned for a mixed-use redevelopment consisting of approximately 14 residential, commercial, and retail buildings totaling roughly 386,500 square feet of conditioned space. The study evaluated the technical, economic, and environmental feasibility of a fifth-generation district geothermal energy system integrated with rooftop solar photovoltaic (PV) to support site-wide electrification and long-term decarbonization.

The feasibility assessment included development of 8760-hour building energy models using Trane TRACE™ 3D+ and simulation of the integrated district energy system using TRNSYS to optimize system sizing, loop temperatures, flow rates, and long-term thermal balance. Multiple geothermal source configurations were evaluated, including vertical closed-loop borefields and optional groundwater heat exchange. The study also assessed conventional all-electric baseline HVAC configurations for comparison and examined modular system phasing strategies to align capital investment with redevelopment timelines.

The recommended system concept utilized a shared ambient thermal loop supporting energy transfer between buildings, enabling modular, phased expansion as redevelopment progressed. The preferred geothermal source consisted primarily of vertical closed-loop boreholes, with provisions for potential future integration of groundwater heat exchange depending on site conditions. Capital cost estimates, energy savings projections, greenhouse gas emissions reductions, and long-term system performance metrics were developed to support investment decision-making.

Role of GEOptimize
GEOptimize served as the geothermal and district energy technical lead. The firm developed detailed building energy models using Trane TRACE™ 3D+, configured and simulated the integrated district system using TRNSYS, designed conceptual ground heat exchanger systems, developed the district

thermal network configuration, and contributed to capital cost estimating, energy savings analysis, and greenhouse gas emissions reduction calculations. The City of Missoula retained ICF as prime consultant, and ICF retained GEOptimize as the geothermal and district energy subconsultant.

Project Cost
Original Contracted Cost: USD 125,000
Final Cost: USD 125,000

Design and Schedule
Anticipated Design Schedule: 7 months
Actual Design Schedule: 7 months

Project Owner
City of Missoula, Montana (with ICF as prime consultant)

Reference Information
Evora Glenn
Community Development Division, City of Missoula
Phone: 406-552-6367
Email: glenne@ci.missoula.mt.us

Craig Schultz
Senior Advisor, ICF
Phone: 202-669-6200
Email: craig.schultz@icf.com

Hopkins Correctional Centre – Ararat, Australia – 2018

Project Description
GEOptimize Inc provided the GHX and district geothermal system design for the Hopkins Correctional Centre in Ararat, Victoria, Australia. The facility consists of a central administration building of approximately 30,000 ft² (2,800 m²) and three groups of inmate cottages totaling approximately 35,000 ft² (3,250 m²), distributed across a 3-acre site.

The administration building is fully heated and air conditioned, while the inmate cottages are served by space heating and domestic hot water only. To reduce CO_2 emissions and avoid fossil fuel use, the Victoria Government selected a GSHP system to provide space conditioning and domestic hot water for the entire facility.

The warm Australian climate and differing building load profiles created a significant thermal imbalance challenge for the vertical borehole GHX systems. The administration building exhibited a cooling-dominant load profile due to high internal gains and year-round cooling demand, resulting in a net annual heat rejection to the ground. Conversely, the cottages were heating-dominant, extracting more heat from the ground annually due to the absence of cooling loads.

If designed as independent GHX systems, the administration borefield would experience long-term temperature rise, reducing heat pump efficiency, while the cottage borefields would experience gradual temperature decline due to sustained heat extraction. Oversizing the borefields would have increased capital cost and only delayed long-term thermal degradation.

To address this, an integrated "energy transfer pipe" system was developed to hydraulically connect the individual GHX modules across the site. This configuration allows thermal energy exchange between the administration building and cottage borefields. Warm fluid from the cooling-dominant administration GHX can be transferred to the heating-dominant cottage GHXs, while cooler fluid from the cottage GHXs can be circulated to the administration building GHX. This district geothermal energy transfer approach moderates loop temperatures across the facility, maintains long-term thermal balance, and preserves heat pump performance without excessive borefield oversizing.

Role of GEOptimize

GEOptimize Inc served as the GHX and district geothermal system designer. The firm developed the vertical borehole GHX designs, engineered the interconnecting district energy transfer piping network, and performed system-level thermal balancing analysis to mitigate long-term ground temperature drift associated with unbalanced building loads.

System installation was performed by Direct Energy Australia. Mechanical system design was completed by NDY Engineering.

Project Cost

Construction Cost: CAD 1,200,000

Design and Schedule

Design Start: March 2017
Construction Completion: January 2019

Project Team

System Installation: Direct Energy Australia
Ground Heat Exchanger and District System Design: GEOptimize Inc.
Mechanical System Design: NDY Engineering

Reference Information

Marcus Wearing-Smith

Director
Direct Energy Australia
Phone: +61 416 736 680
Email: marcus@directenergy.com.au

Max Ploumis
Engineering Manager – Renewable Energy & Environmental Engineer
Direct Energy Australia
Phone: +61 431 212 740
Email: max@directenergy.com.au

Section D – Experience of Key Personnel Assigned to the Project

The feasibility study will be delivered by a focused project team consisting of four key personnel with complementary expertise in energy modelling, GHX design, district system simulation, and technical oversight. The team structure is designed to ensure that senior leadership remains directly involved throughout the study while day-to-day analytical tasks are executed by specialists with relevant project experience.

Ian Lohrenz will be responsible for the development of building energy models to quantify hourly heating and cooling loads and establish baseline and proposed system performance scenarios.

Stephen Hak will lead the design and analysis of the district energy system, utilizing TRNSYS to evaluate subsurface thermal performance, system configurations, and long-term thermal balance. PIPE-FLOW will be used for fluid dynamics analysis of the system.

Connor Dacquay and **Ed Lohrenz** will provide overall technical oversight and quality control for the study. Their roles include guiding system configuration decisions, reviewing modelling assumptions and results, coordinating inputs across disciplines, and ensuring alignment with feasibility study objectives and RFP requirements.

All key personnel will contribute to the development, review, and finalization of the feasibility study report. An organizational chart illustrating team roles is provided below.

Ed Lohrenz - Founder
GSHP system specialist & industry pioneer

Connor Dacquay
Geothermal Engineer
GSHP system specialist

Ian Lohrenz
Geothermal Engineer
Energy modelling specialist

Stephen Hak
Geothermal Engineer
District system modelling specialist

Key Personnel

Ed Lohrenz, BES, CGD

Ed has worked in the GSHP industry since the early 1980s and brings over four decades of experience in the design and implementation of geothermal energy systems. Throughout his career, he has collaborated with leading practitioners in the field and has contributed to projects across a wide range of applications and climatic conditions, from the Australian outback to the Siberian tundra, as well as numerous commercial and institutional projects throughout Canada and the United States.

Ed was a key contributor to the development of the Certified Geo-Exchange Designer (CGD) course for the International Ground Source Heat Pump Association (IGSHPA), which is recognized as the industry standard for commercial GSHP system design.

Comparable Project 1

Canadian Mennonite University – District GSHP System

Role of Ed Lohrenz:
Principal and senior technical advisor providing system design oversight, phasing strategy development, review of GHX configuration and installation approach, and guidance on monitoring and long-term system performance optimization.

Comparable Project 2

Water Tower District – District GSHP Feasibility Study

Role of Ed Lohrenz:
Principal-in-Charge and technical lead responsible for system concept development, review of energy modelling assumptions, evaluation of geothermal source options (open-loop, closed-loop, and hybrid

configurations), and oversight of feasibility, risk, and cost optimization analysis.

Stephen Hak, EIT, BSc

Stephen Hak is a Biosystems Engineering graduate from the University of Manitoba and is currently pursuing a Master of Science in Engineering with a focus on geothermal energy systems. He has specialized experience in the modelling, design, and analysis of GSHP and district-scale geothermal systems, with particular expertise in the use of TRNSYS for simulating ambient-temperature, multi-building thermal energy networks and has been working in the GSHP industry for six years.

Stephen has completed more than ten (10) district GSHP feasibility studies, contributing to energy modelling, GHX design, system configuration analysis, and feasibility reporting for complex, multi-building projects.

Comparable Project 1

Community of Inverness – District GSHP Feasibility Study

Role of Stephen Hak:
Lead engineer responsible for conceptual district GSHP system design, TRNSYS-based system modelling, evaluation of alternative heat source configurations, and preparation of the feasibility study report outlining technical, economic, and environmental outcomes.

Comparable Project 2

University of Minnesota Morris – District GSHP Feasibility Study

Location: Morris, Minnesota, USA
Project Owner: University of Minnesota Morris
Client: Otter Tail Power Company

Project Description:
Feasibility study evaluating the conversion of approximately 28 campus buildings from a central boiler and chiller system to a district GSHP system. The study assessed the integration of open-loop groundwater heat exchangers with horizontally directionally drilled closed-loop ground heat exchangers, with a focus on reducing carbon emissions and long-term energy costs.

Role of Stephen Hak:
District GSHP system designer responsible for system configuration development, energy modelling,

economic and lifecycle cost analysis, greenhouse gas reduction assessment, and preparation of the final feasibility study report.

Contact: Jon Fabre, Otter Tail Power Company, 218-739-8633, jfabre@otpco.com

Ian Lohrenz, P.Eng, CGD
Ian Lohrenz is a licensed professional engineer with a Bachelor of Science in Mechanical Engineering from the University of Manitoba (2008). His professional practice has focused on building energy efficiency, low-carbon mechanical systems, and the application of GSHP technology in both single-building and district-scale contexts for the past 18 years.

Ian has developed specialized expertise in building energy modelling, GHX design support, GSHP systems, building commissioning, and energy auditing. His role on feasibility and retrofit studies typically involves developing detailed and calibrated energy models to quantify baseline performance, assess alternative system configurations, and support technical and economic decision-making.

Comparable Project 1

Missoula Triangle District Energy System (DES) Study

Role of Ian Lohrenz:
Lead building energy modeller responsible for developing detailed hourly energy models to quantify building loads, support GSHP system sizing, and evaluate the impact of rooftop solar generation on overall system performance.

Comparable Project 2

Bethel Place – Deep Energy Retrofit Feasibility Study

Location: Winnipeg, Manitoba
Project Owner: Bethel Mennonite Care Services

Project Description:
Feasibility study examining multiple deep energy retrofit options for a seniors' apartment building, including conversion from an existing boiler-based heating system to a GSHP system. The study was undertaken by a multidisciplinary local project team and focused on reducing energy consumption, emissions, and long-term operating costs.

Role of Ian Lohrenz:
Energy audit and modelling lead responsible for completing an ASHRAE Level 2 Energy Audit, developing calibrated baseline and proposed energy models, supporting GHX sizing, and quantifying anticipated energy savings and performance improvements.

Contact: Jim Nolsted, SEEFAR Building Analytics, 431-373-8255, info@seefar-valuation.com

Connor Dacquay, Dr. Eng, P.Eng, CGD
Connor Dacquay is a licensed professional engineer in the Provinces of Manitoba and Ontario with extensive experience in district-scale GSHP systems, monitoring, feasibility studies, and detailed engineering design. He has contributed to the planning and implementation of large-scale heat pump systems across North America, Asia, and Australia and has been involved in the GSHP industry for over 8 years.

Connor is a Certified GeoExchange Designer (CGD) and instructor, combining advanced technical expertise with a strong focus on education, quality assurance, and knowledge transfer. His professional practice spans feasibility analysis, system optimization, lifecycle performance evaluation, and transition of district geothermal systems from concept through detailed, constructible design.

Comparable Project 1

Bemidji State University – District GSHP Feasibility Study

Role of Connor Dacquay:
Technical lead responsible for oversight of feasibility methodology, review and validation of calibrated 8760-hour energy models, evaluation of system configurations, assessment of capital cost and lifecycle economics, and integration of energy efficiency measures critical to system viability. Connor guided system selection toward an ambient "energy transfer pipe" (ETP) district configuration to maximize load sharing and reduce GHX infrastructure requirements.

Comparable Project 2

Ithaca Urban Thermal Energy Network (UTEN) – District Geothermal System Design

Location: Ithaca, New York, USA
Project Owner: Avangrid

Project Description:
Detailed engineering design of a district geothermal energy system serving 44 mixed-use buildings as part of a proposed Urban Thermal Energy Network (UTEN). The project used an ambient-loop district system from feasibility into full engineering design, integrating multiple geothermal heat sources and sinks to support long-term scalability and energy balance.

Role of Connor Dacquay:
Senior technical lead responsible for district geothermal system design, including validation of building energy models, evaluation of geological and hydrogeological conditions, advanced TRNSYS-based system simulation, and optimization of long-term thermal balance. Connor oversaw the design of the ambient loop infrastructure, GHX and heat source/sink integration, hydronic system sizing, and development of control and monitoring strategies to ensure lifecycle performance and resilience.

Contact: Austin Crosby, Labella Associates, 716-710-3058, acrosby@labellapc.com

Section E – Project Understanding and Methodology

Project Understanding

The City of Winnipeg is exploring a district thermal energy system along Graham Avenue as an opportunity to reduce greenhouse gas emissions, improve energy efficiency, and support the revitalization of downtown. With buses recently rerouted off Graham Avenue, the street's planned reconstruction creates a timely opening to install thermal infrastructure beneath the roadway and sidewalks. The vision is to connect multiple buildings on or near Graham Avenue into a shared heating and cooling network that can recover waste heat from buildings that have excess and distribute it to those with heating demand, supplemented as required by ground heat exchangers or other thermal sources and sinks to balance loads and provide seasonal energy storage.

The project area includes several major facilities with committed or potential interest in the system. Key stakeholders identified by the City include: Southern Chiefs Organization (450 Portage), Women's Health Clinic (419 Graham), Royal Winnipeg Ballet (380 Graham), Manitoba Hydro (360 Portage), True North Real Estate &/or Longboat (236 Carlton, 242 Hargrave, 260 Hargrave, Canada Life Centre), Manitoba Public Insurance (333 St. Mary), and City of Winnipeg buildings (251 Donald Street, 266 Graham). Many of these buildings have significant heating and cooling demands and are in close proximity along Graham Avenue, making them prime candidates for a thermal network (Figure 1). Their participation could allow the district system to aggregate loads, improve overall efficiency through diversity of demand, and share both costs and benefits of the new infrastructure.

The policy context for this project is strongly supportive. Winnipeg's Climate Action Plan calls for innovative approaches to reduce building-related greenhouse gas emissions, which represent a significant portion of city-wide emissions, with nearly half attributable to natural gas heating. A district thermal energy system on Graham Avenue directly addresses these goals by cutting reliance on fossil-fuel heating. Additionally, the downtown development vision (CentrePlan 2050) and Complete Communities 2.0 strategy encourage sustainable infrastructure investments that make the downtown more liveable and attractive. This feasibility study is funded in part by external grants (Federation of Canadian Municipalities and Efficiency Manitoba), indicating broad support for exploring a landmark project that could become a showcase for urban sustainability in Winnipeg.

In summary, our team understands that the City is seeking a comprehensive feasibility study and business case that will answer the following key questions:

- *Technical Feasibility:* What is the most viable conceptual design for a Graham Avenue district thermal energy system, considering local geological conditions, existing building systems, and available technologies (geothermal closed-loop vs open-loop, heat recovery systems, thermal storage, etc.)? What infrastructure (boreholes, pipes, energy centers) would be required, and where could it be located given the urban context (accounting for underground constraints like service tunnels or areaways beneath sidewalks)? How would the system be controlled and operated to reliably meet heating and cooling demands?

- *Energy Performance and GHG Impact*: How does the proposed district thermal energy system perform relative to a business-as-usual scenario in which each building operates independent, conventional HVAC systems? The study will quantify changes in annual energy consumption, peak load demand, and greenhouse gas emissions at both the district and individual building levels. A key objective is to clearly demonstrate the system's potential contribution to the City's climate goals, including estimated annual and lifecycle (25-year) carbon emission reductions.

- *Economic Feasibility:* What are the anticipated capital and operating costs of implementing and operating the district thermal energy system, and how do these compare to a business-as-usual scenario? The study will develop Class C–level capital cost estimates, including appropriate contingencies, for all major system components, along with projected operating and maintenance costs and potential revenues where a utility rate structure is considered. A lifecycle financial analysis will be performed to evaluate key metrics such as Net Present Value, Internal Rate of Return, and payback period, clearly defining the conditions under which the project is financially viable for the City and participating stakeholders, including the influence of grant funding, energy price escalation, and potential carbon pricing.

- *Implementation and Risk:* What technical, financial, and regulatory risks are associated with the proposed district thermal energy system, and how can they be effectively mitigated? The study will identify key risks such as long-term ground temperature performance, integration with existing building systems, extreme cold backup requirements, energy price volatility, and construction cost uncertainty, and will propose practical mitigation strategies for each. Based on the results, a clear go/no-go decision framework will be developed to support the City's determination of whether to proceed to detailed design and implementation. If the project is deemed viable, the study will outline recommended next steps, including targeted field testing, detailed design, procurement strategy, and phasing considerations; if not, it will identify alternative or incremental measures to advance downtown energy efficiency and emissions reduction objectives.

- *Stakeholder and Organizational Considerations:* How should the district thermal energy system be structured in terms of ownership, operation, and cost-sharing among participating parties? The study will examine potential governance and delivery models, such as City-owned utility operation, third-party ownership, or public–private partnership arrangements, and assess their implications for risk allocation, cost recovery, and long-term management. The study will also address how future developments and building retrofits can be designed to integrate with the district system, including the preparation of preliminary developer guidelines to support connection readiness, such as mechanical room allowances, heat exchanger interfaces, and HVAC system compatibility.

Our understanding is that the deliverables will include a geotechnical desktop report, a conceptual design package, energy modeling results for both the proposed system and baseline, interim progress reports, and draft and final feasibility reports. All these will be prepared in collaboration with City staff and Efficiency Manitoba, with regular in person or virtual review meetings to ensure alignment. Our team's direct experience with Winnipeg subsurface conditions, regulatory processes, and local utility coordination reduces implementation uncertainty relative to out-of-province consultants. Ultimately, our aim is to provide the City with a decision-enabling document: a feasibility study that is thorough, transparent, and tailored to Winnipeg's specific context, giving City Council and stakeholders the evidence they need to move forward confidently.

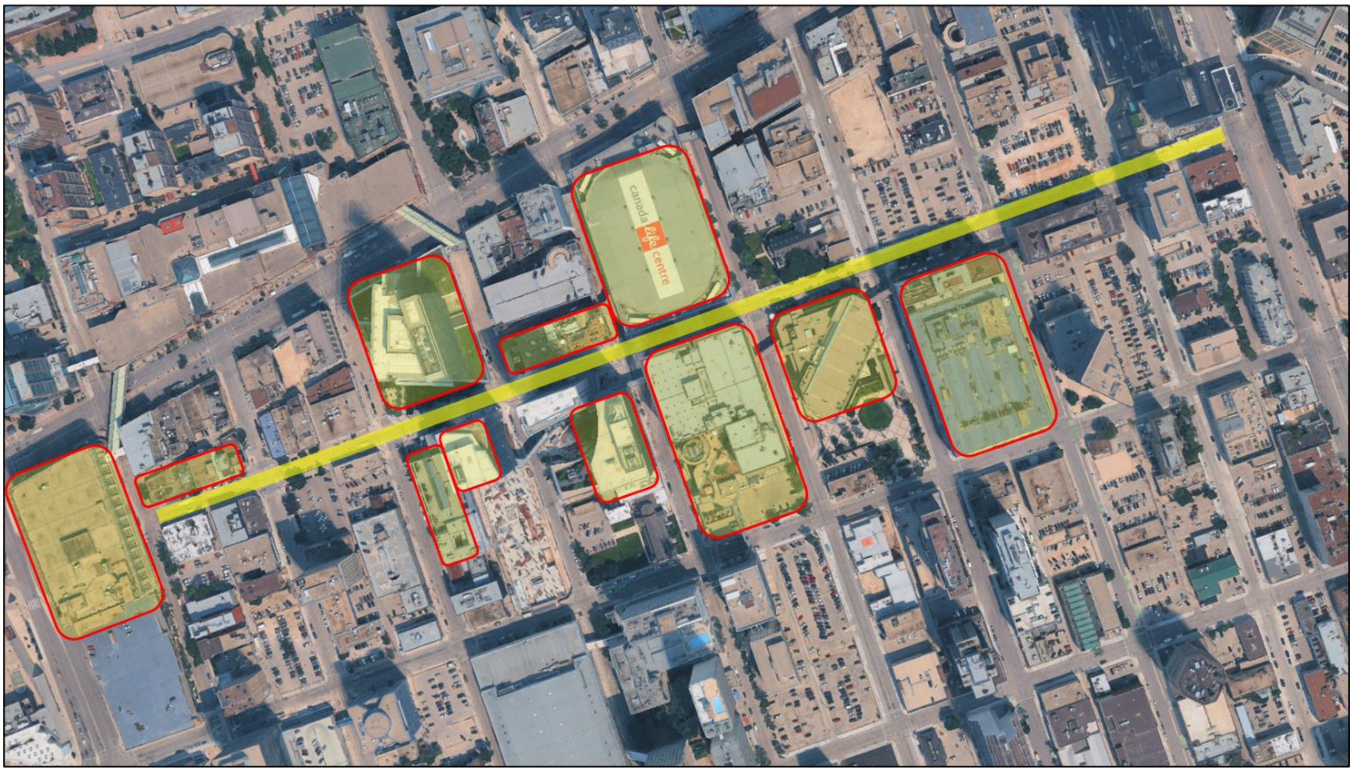


Figure 1: Graham Avenue study area and participating buildings.

Methodology

Our methodology combines technical analysis with a collaborative, stakeholder-informed approach. We will execute the project in sequential phases aligned with the City's required deliverables and decision milestones. Below, we outline each phase of work, including its objectives, approach, and deliverables. This phased approach ensures all RFP requirements are met or exceeded, and it provides clear checkpoints for City input and feedback.

Phase 1: Project Start-Up Meeting

Objective:
Initiate the project by confirming scope, assumptions, data requirements, roles, and communication protocols, ensuring a shared understanding among the City's Project Team, Efficiency Manitoba, and the consultant team.

Approach:
GEOptimize will review all background information provided and prepare a focused agenda covering the work plan, schedule, roles and responsibilities, and initial data requirements. An initial Project Start-Up Meeting (in-person in Winnipeg where possible, or virtual) will be held with the City Project Team and Efficiency Manitoba, with participation from other stakeholders as appropriate.

The meeting will be used to:

- Confirm project objectives, scope boundaries, deliverables, and milestones.
- Establish roles, reporting relationships, communication protocols, and document control.
- Confirm the business-as-usual baseline definition for comparative analysis.
- Review building data requirements, utility data availability, and points of contact.
- Confirm performance targets, evaluation criteria, and economic analysis assumptions.
- Review the project schedule, review cycles, and any fixed deadlines.
- Discuss site access, safety considerations, and stakeholder engagement approach.

Following the meeting, GEOptimize will document decisions and action items, and issue a finalized data request list to guide subsequent phases.

Deliverable:
Project Start-Up Meeting agenda and a finalized data request list.

Phase 2: Desktop Geotechnical Study

Objective:
Complete a desktop geotechnical and hydrogeological assessment of the Graham Avenue corridor to characterize subsurface conditions relevant to geothermal system feasibility, identify constraints and opportunities for GHX installation, inform the selection of closed-loop versus open-loop system concepts, and recommend any physical testing required to reduce uncertainty prior to implementation.

Approach:
GEOptimize will compile and review available subsurface information for the study area, including borehole logs, foundation reports, City infrastructure records, and regional geological mapping. The review will focus on soil and bedrock conditions, depth to bedrock, groundwater levels, and the presence of aquifers. Existing underground features such as utility corridors, parkades, and building areaways will be identified and mapped where information is available, recognizing the unique subsurface connectivity along Graham Avenue.

Based on the compiled information, GEOptimize will estimate key thermal parameters required for feasibility analysis, including ground thermal conductivity, thermal diffusivity, and undisturbed ground temperature. These values will be informed by local geology and available site data and will be used to support preliminary GHX sizing and long-term performance assessment. Where available, operating data from nearby geothermal systems (such as Manitoba Hydro Place) will be reviewed to refine assumptions. In addition, the GEOptimize team already possesses thermal conductivity test results from three locations within approximately 2 km of the proposed study area. These data provide valuable local validation of subsurface thermal properties and will be incorporated into the analysis. Figure 2 illustrates the locations and results of these tests relative to the project site.

The desktop study will also assess the feasibility of groundwater-based (open-loop) geothermal options by reviewing available well records and hydrogeological data to determine whether sufficient groundwater yield, quality, and reinjection potential exist. Water quality considerations and regulatory implications will be noted to inform conceptual design decisions.

The assessment will identify key constraints and risks, including constructability limitations, space availability for drilling, potential contamination concerns, groundwater-related risks, and long-term thermal balance considerations. Coordination with Public Works and available corridor design information will inform realistic assumptions regarding feasible drilling locations within the right-of-way or adjacent properties.

Deliverables:
GEOptimize will prepare a concise Desktop Geotechnical Report summarizing:

- Expected geological and hydrogeological conditions.
- Implications for GHX feasibility and concept selection.
- Key risks and uncertainties affecting long-term system performance.
- Recommendations for physical testing (e.g., test boreholes, thermal response testing, well pump testing), including indicative budgetary costs.

The findings will be reviewed with the City Project Team and Efficiency Manitoba to confirm assumptions and inform the subsequent conceptual design and modelling phases.

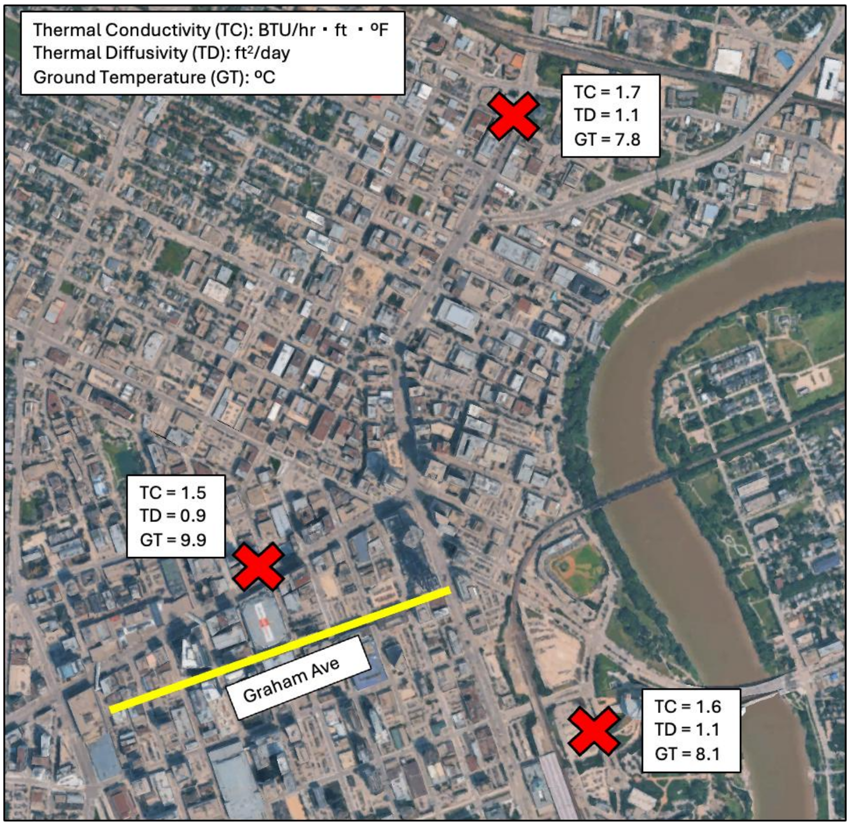


Figure 2: Three thermal conductivity test results within 2 km of the proposed project site.

Phase 3: Preliminary Progress Report & Review

Objective:
At the mid-point of the study, provide the City with a structured interim progress report summarizing preliminary energy modelling results, emerging system concepts, and key assumptions, enabling early validation of direction and allowing for mid-course adjustments before advancing to detailed conceptual design and business case development.

Approach:

Preliminary Energy Modelling
By this stage, GEOptimize will have developed preliminary building energy models for the primary buildings included in the study using Trane TRACE™ 3D+ and complementary analysis tools (Figure 3). These 8760 models will represent each building's annual heating and cooling performance under a business-as-usual and post-retrofit configuration, based on available architectural, mechanical, and operational data.

Model calibration will be applied selectively on a building-by-building basis, depending on the availability of utility data and the relative importance of each building to overall district system performance. Buildings that are expected to represent major thermal sources or sinks, or that materially influence district system sizing, will receive a higher level of calibration where sufficient data is available. Conceptual-level models will be used for buildings where detailed data cannot be obtained, while still providing reliable load estimates for feasibility-level analysis.

An initial district system representation will also be developed to evaluate load aggregation, energy-sharing potential, and preliminary system configuration options. Early modelling will focus on the highest-impact buildings within the confirmed study area, including major institutional and arena facilities, to establish governing heating and cooling loads and inform preliminary distribution system sizing. The analysis will assess opportunities for internal energy sharing, use of existing cooling and heat rejection systems, and alternative strategies to improve thermal balance and reduce reliance on ground heat exchanger infrastructure.

The preliminary modelling results will include:

- Building descriptions and modelling assumptions, including floor area, envelope characteristics, occupancy profiles, ventilation rates, and existing or planned HVAC system efficiencies, to confirm accuracy with the City and participating stakeholders.
- Baseline versus district load profiles, summarizing annual heating and cooling energy use and peak demands for each building and the district as a whole. The analysis will identify heating- and cooling-dominant buildings and evaluate the degree of thermal diversity and load balancing achievable within the corridor.
- Preliminary thermal balance assessment, identifying whether the combined system is heating- or cooling-dominated and the resulting implications for ground heat exchanger sizing and the need for auxiliary heat rejection or injection.

Integration of Geotechnical Findings

Key findings from the desktop geotechnical and hydrogeological study will be integrated into the emerging concept direction. Any constraints related to drilling feasibility, space availability, or groundwater considerations that influence system configuration will be highlighted, and the continued viability of closed-loop, open-loop, or hybrid concepts will be confirmed.

Emerging Concept Design Direction

The progress report will outline the emerging district system concept at a high level, including:

- System configuration, such as heat recovery options, use of buffer storage tanks and energy-sharing strategies, closed-loop vertical boreholes, hybrid approaches, or groundwater-based systems, and their indicative locations within or adjacent to the Graham Avenue corridor.
- Energy plant strategy, including centralized versus decentralized heat pump arrangements and the rationale for the preferred approach.

- Auxiliary components, such as backup heating for extreme cold conditions, heat rejection equipment for thermal balance, and any preliminary consideration of thermal storage.
- High-level controls and operations concept describing temperature management of the thermal network through monitoring and data acquisition.
- Confirmation of baseline systems, ensuring agreement on the conventional systems used for comparative analysis.

Preliminary Economic Insights

While detailed financial analysis will be completed in later phases, the progress report will provide an early indication of:

- Order-of-magnitude capital cost ranges for major system components.
- Directional operating cost impacts resulting from fuel switching and electrification.
- Identification of obvious economic opportunities or constraints to be examined further in the business case.

Deliverable

The above findings will be compiled into a concise Preliminary Progress Report, including narrative summaries, preliminary data tables, and schematic illustrations as appropriate.

GEOptimize will then facilitate a Progress Review Meeting with the City Project Team and Efficiency Manitoba to:

- Validate modelling assumptions and preliminary results.
- Confirm alignment with City and stakeholder expectations.
- Identify any required refinements, additional scenarios, or changes in scope prior to advancing to detailed conceptual design and modelling.

Meeting outcomes and agreed adjustments will be documented and incorporated into subsequent phases.

Figure 3: Example Trane TRACE™ 3D+ 8760 energy model for a GSHP system design.

Phase 4: Conceptual Design & Energy Modelling

Objective:
Develop a detailed conceptual design for the district thermal energy system and complete comprehensive 8760 energy modelling for both the proposed system and the business-as-usual baseline. This phase produces the core technical outputs of the study, a refined system concept supported by advanced performance modelling, to inform financial analysis, risk assessment, and final decision-making.

Approach:

Conceptual District System Design
Building on the confirmed direction from Phase 3, GEOptimize will prepare a conceptual design package defining the physical configuration and operation of the district thermal energy system, including:

- Site layout diagram illustrating GHX locations, distribution piping routes along Graham Avenue, building connection points, and centralized or distributed energy plant locations, with consideration of constructability constraints and underground infrastructure.
- Hydronic system schematic showing building interfaces, heat pump arrangements, circulation loops, auxiliary equipment, and key control elements to clearly illustrate energy flows and operating temperature ranges. Industry standard hydraulic modelling software PIPE-FLOW will be used to calculate necessary flow rates, pressure drops, and Reynolds numbers.
- Conceptual equipment schedule identifying preliminary quantities and capacities of major system components sufficient to support Class C cost estimating.
- Controls and operations narrative describing seasonal operating modes, load sharing, loop temperature management, prioritization of ground-source and recovered heat, and the role of auxiliary systems for resiliency.

Advanced Energy Modelling Using TRNSYS

GEOptimize will employ TRNSYS as the primary district system simulation platform to evaluate integrated system performance over time (Figure 4). TRNSYS enables fully coupled, transient modelling of buildings, GHXs, heat pumps, auxiliary equipment, controls, and thermal interactions, providing a level of analytical rigor not achievable using spreadsheet-based methods or single-purpose sizing tools alone.

Finalized 8760-hour heating and cooling load profiles for each participating building will be integrated directly into the TRNSYS district energy model to support detailed system-level simulation. The building energy models will also be used to quantify building-level energy consumption, operating costs, and greenhouse gas emissions for both the business-as-usual scenario and the ground-source district energy system option, and to calculate Thermal Energy Demand Intensity (TEDI) and Energy Use Intensity (EUI) metrics.

Scenario Analysis and Performance Outputs

The modelling framework will evaluate, at minimum:

- A business-as-usual baseline scenario, with independent conventional HVAC systems.
- A district system scenario, with all participating buildings connected to the conceptual design.

Additional scenarios such as alternative system configurations or participation levels will be evaluated as directed by the City.

Key outputs will include:

- Annual energy consumption by source and peak heating and cooling capacities.
- Greenhouse gas emissions and reductions, expressed annually and over the system lifecycle.
- Renewable energy fraction and degree of energy sharing between buildings.
- Long-term GHX performance, including projected ground temperature trends.
- Auxiliary equipment type and size based on thermal balancing and/or peak heating or cooling requirements.
- Seasonal thermal energy storage behaviour within the ground.
- Electrical demand impacts and implications for utility coordination.

All modelling assumptions, inputs, and weather data will be documented to ensure transparency and reproducibility.

Deliverable
GEOptimize will compile the results into a Conceptual Design and Energy Modelling Report and present the findings at an in-person or virtual Conceptual Design Review Meeting with the City Project Team and Efficiency Manitoba. The review will confirm the preferred concept and modelling results as a sound technical basis for the subsequent financial analysis, risk assessment, and final feasibility reporting.

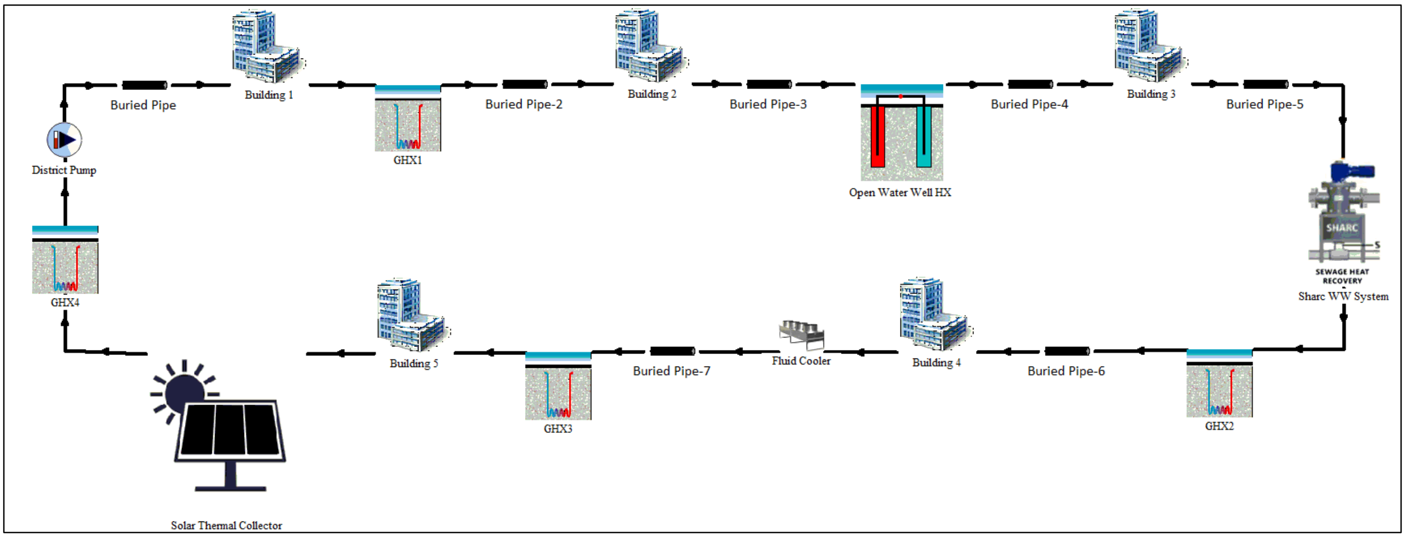


Figure 4: Example TRNSYS Model.

Phase 5: 75% Draft Feasibility Report & Review

Objective:
Compile all technical, financial, and environmental findings into a comprehensive draft feasibility study report (approximately 75% complete) to support City and stakeholder review. This phase provides a structured opportunity to validate assumptions, results, and conclusions prior to finalization, ensuring the final report is complete, accurate, and decision-ready.

Approach:
GEOptimize will prepare a structured draft report that documents the full feasibility assessment in a clear and transparent manner. The draft report will include:

- ***Introduction and Background***
  Summary of project context, objectives, scope, and policy drivers, including the role of the study in supporting the City's climate and energy transition goals.

- ***Site and Geotechnical Summary***
  Key findings from the desktop geotechnical and hydrogeological study, highlighting how subsurface conditions influenced system feasibility and conceptual design decisions. Detailed

supporting material will be included in appendices as appropriate.

- ***Methodology***
  Description of the analytical approach, tools, and assumptions used throughout the study, including building energy modelling, district system simulation, and financial analysis methodologies. This section will document modelling software (e.g., Trane TRACE™ 3D+, PIPE-FLOW, and TRNSYS), climate data sources, and financial assumptions to ensure transparency and reproducibility.

- ***Conceptual Design Description***
  Detailed narrative of the preferred district thermal energy system concept as finalized in Phase 4, including system configuration, major components, and operating strategy, with supporting diagrams and equipment schedules. This section will clearly describe the proposed system relative to the business-as-usual baseline and outline anticipated next steps.

- ***Energy and Environmental Analysis***
  Presentation of full modelling results for baseline and district system scenarios, including annual energy consumption, peak demands, Energy Use Intensity (EUI), Thermal Energy Demand Intensity (TEDI), greenhouse gas intensity, and total emissions reductions. Results will be summarized at both the building and district levels and expressed in clear, decision-oriented metrics.

- ***Financial Analysis and Business Case***
  Development of a feasibility-level business case including:
  - Class C capital cost estimates for system implementation, broken down by major cost categories and supported by stated assumptions.
  - Estimated annual operating and maintenance costs compared to the baseline scenario.
  - Lifecycle cash flow analysis over the selected study period, including equipment replacement and energy price escalation.
  - Financial performance metrics such as Net Present Value, Internal Rate of Return, levelized cost of thermal energy, and simple payback.
  - Sensitivity analysis examining the impact of key variables such as energy prices, participation levels, and capital cost uncertainty.
  - High-level discussion of ownership, operation, and cost-recovery considerations, including potential utility or partnership models.

- ***Risk and Resilience Assessment***
  Identification of technical, financial, regulatory, and organizational risks, along with proposed mitigation strategies. This will include consideration of long-term thermal performance, system

integration risks, cost uncertainty, stakeholder participation, and climate resilience factors.

- ***Preliminary Go/No-Go Recommendation***
  A clear, defensible recommendation regarding whether the City should proceed to implementation, including the conditions required for success. Where appropriate, recommended next steps such as further testing, detailed design, procurement strategy, and phasing will be outlined. If the project is not recommended to proceed, alternative or incremental actions to advance downtown energy sustainability will be identified.

- ***Draft Developer Guidelines***
  Preliminary guidance for future developments and major retrofits within the study area to facilitate potential future connection to a district thermal energy system, including considerations for mechanical space planning, interface requirements, and HVAC system compatibility.

Deliverables

The 75% Draft Feasibility Study Report will be issued to the City Project Team and Efficiency Manitoba for review. GEOptimize will support the review process through an in-person or virtual Draft Report Review Meeting to present key findings, discuss feedback, and confirm required revisions. All comments and direction received will be documented and incorporated into the final feasibility report.

Phase 6: Final Feasibility Report & Presentation

Objective:

Produce the final feasibility study report incorporating all feedback received during the draft review and present the findings to the City and key stakeholders, concluding the project with a clear recommendation and a defined path forward.

Approach:

Final Feasibility Report Completion

GEOptimize will revise the draft feasibility study to address all comments and direction provided by the City Project Team and Efficiency Manitoba. The Final Feasibility Study Report will be a polished, comprehensive document suitable for distribution to decision-makers and funding partners.

The final report will:

- Clearly emphasize the City's preferred findings and recommendations, as directed during the draft review.
- Confirm that all RFP-required content has been addressed, including technical analysis, cost estimates, governance considerations, and risk assessment.

- Incorporate an executive summary for senior leadership and external audiences.
- Include supporting appendices containing detailed technical information such as conceptual design drawings, cost estimate back-up, energy modelling outputs, and stakeholder consultation summaries, allowing the main report body to remain concise and decision-oriented.

Recommendation and Implementation Roadmap

The final report will clearly state a go/no-go recommendation, supported by technical, financial, and environmental analysis.

If a "go" decision is recommended, the report will outline a high-level implementation roadmap, including:

- Recommended next steps such as targeted field testing, detailed engineering design, funding and partnership development, procurement strategy, and commissioning and monitoring considerations.
- Phasing strategies to allow staged implementation, where appropriate, based on building readiness, capital availability, or corridor reconstruction timing.

If a "no-go" recommendation is provided, the report will explain the rationale and identify alternative or incremental measures to advance downtown energy efficiency and emissions reduction objectives. Where appropriate, recommendations for additional investigation to reduce uncertainty will be included.

Final Presentation and Close-Out

GEOptimize will deliver a final presentation in-person to the City Project Team, Efficiency Manitoba, and other attendees invited by the City. The presentation will summarize key findings, the recommended system concept, costs, benefits (including greenhouse gas reductions), identified risks, and next steps, and will provide an opportunity for discussion and clarification as the City moves toward a decision.

Following delivery of the final report and presentation, GEOptimize will remain available to respond to follow-up questions or provide clarification to support internal review or briefing of senior decision-makers.

Deliverables

Completion of Phase 6 provides the City with a fully documented feasibility study and business case that satisfies all requirements of the RFP and supports informed decision-making regarding implementation of a district thermal energy system along Graham Avenue.

Quality Assurance and Innovation

Throughout all phases, GEOptimize will apply a strong quality control process. Key calculations and assumptions will be double-checked internally. We will also ensure all City and Efficiency Manitoba requirements are met: for instance, aligning report format or including any specific metrics they need for their purposes. We will leverage innovative practices to provide the City with creative yet practical solutions. Our team stays updated on global best practices in district energy knowledge from European low-temperature district heating networks and Canadian examples in Ottawa and Vancouver, from which relevant lessons can be drawn.

Finally, stakeholder engagement is woven into our methodology at every stage: from data collection to design review to final recommendation, we will ensure the City and other key players have input and buy-in. This collaborative approach not only improves the study with local insights but also paves the way for smoother implementation if the project moves forward, as stakeholders will already be familiar with and supportive of the concept.

Section F - Project Schedule

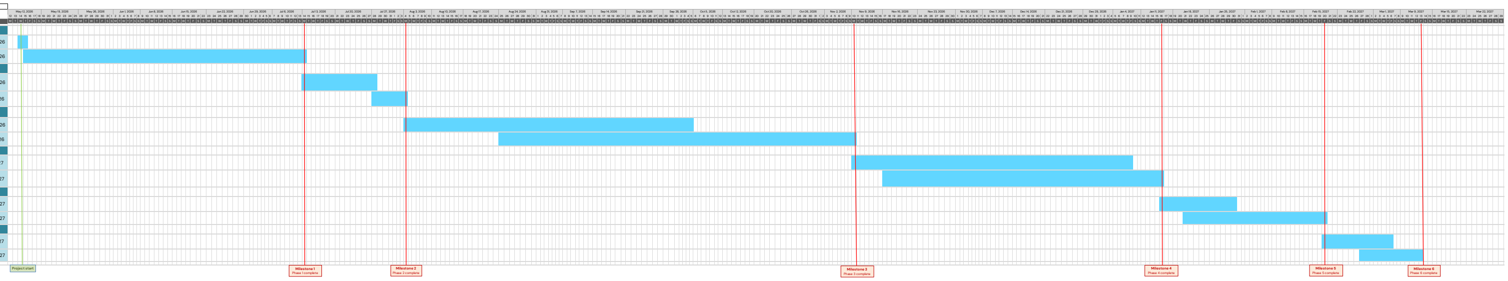
We propose the following schedule for the feasibility study, structured to meet the City's needs and critical milestones. The timeline is aligned with the critical milestones outlined in the RFP and assumes a notice of award by May 13, 2026. The project duration is approximately 11 months from kickoff to final report, ensuring thorough analysis and stakeholder engagement while delivering results in a timely manner. The major milestones are as follows:

- Project Start Meeting – May 20, 2026
- Geotechnical Report – August 4, 2026
- Preliminary Progress Report & Meeting – November 10, 2026
- Conceptual Design & Energy Modelling – January 16, 2027
- Draft (75%) of Feasibility Report – February 20, 2027
- Final Feasibility Report – March 13, 2027

For more information, please see the Gantt chart on the following page.

Start Date
2026-05-13

TASK	ASSIGNED PERSONNEL	DAYS	START	END
Phase 1: Project Start-Up Meeting & Information Gathering				
Kick-off meeting	SH/IL/CD/EL	1	5-13-26	5-14-26
Information gathering	SH/IL/CD	60	5-14-26	7-13-26
Phase 2: Desktop Geotechnical Study				
Reviewing available subsurfance information for the study area	SH/IL/CD	14	7-13-26	7-27-26
Development of geotechnical report	SH/IL/CD	7	7-27-26	8-3-26
Phase 3: Preliminary Progress Report & Review				
Preliminary 8760 building energy models development	IL	62	8-3-26	10-4-26
Preliminary district design & option analysis	SH/EL/CD/IL	77	8-24-26	11-9-26
Phase 4: Conceptual Design & Energy Modelling				
Refine 8760 building energy models	IL	60	11-9-26	1-8-27
Firming up conceptual district system design using feedback from stakeholders	SH/EL/CD/IL	60	11-16-26	1-15-27
Phase 5: 75% Draft Feasibility Report & Review				
Economic & energy use analysis including Class C pricing	SH/IL/CD/EL	14	1-15-27	1-29-27
Report writing	SH/IL/CD/EL	30	1-20-27	2-19-27
Phase 6: Final Feasibility Report & Presentation				
Final report writing	SH/IL/CD/EL	14	2-19-27	3-5-27
Final presentation	SH/IL/CD/EL	14	2-26-27	3-12-27



**Stephen Hak = SH, Ian Lohrenz = IL, Connor Dacquay = CD, Ed Lohrenz = EL

GEOptimize

167 Lombard Avenue, Suite 500
Winnipeg, MB R3B 0V3
+1 (204) 996-8133